

Increasing Logarithmic Weights for Rolling Validation: Theory

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1 Setting and goal

This report studies increasing logarithmic-type weights for rolling validation in a Gaussian oracle-versus-mean comparison. The goal is to understand how increasing weights affect both the expected rolling-validation signal and its variability, as a step toward useful growth conditions for weighting sequences. Decreasing weights are outside the current research direction. Finite-sample behavior is studied separately in the accompanying report, *Increasing Logarithmic Weights for Rolling Validation: Simulation Study* (08B).

Let $Y_i = \theta_\star + \xi_i$, where $\xi_i \stackrel{\text{iid}}{\sim} N(0, 1)$. At validation time i , compare the oracle $\hat{\theta}_{i-1}^{(1)} = \theta_\star$ with the fitted mean

$$\hat{\theta}_{i-1}^{(0)} = \frac{1}{i-1} \sum_{t=1}^{i-1} Y_t.$$

For positive deterministic weights w_i , define

$$\Xi_n = \sum_{i=2}^n w_i \left[(Y_i - \hat{\theta}_{i-1}^{(0)})^2 - (Y_i - \hat{\theta}_{i-1}^{(1)})^2 \right]. \quad (1)$$

A positive value favors the oracle. Thus $\Xi_n < 0$ is incorrect selection, and selection consistency means $\mathbb{P}(\Xi_n < 0) \rightarrow 0$.

1.1 Core quantities and selection criterion

Expected signal. For any positive deterministic weights,

$$\mathbb{E}[\Xi_n] = \sum_{i=2}^n \frac{w_i}{i-1}. \quad (2)$$

This is the accumulated deterministic advantage of the oracle under the chosen weighting sequence.

Variability. Successive validation terms share training observations, so their dependence must be retained. The exact variance decomposition and its Gaussian orthogonality argument are given in Appendix A.

Chebyshev selection rule. Since $\mathbb{E}[\Xi_n] > 0$,

$$\mathbb{P}(\Xi_n < 0) \leq \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2}. \quad (3)$$

Accordingly, each candidate weight is treated in three steps: derive the expectation, derive the variance, and check whether the Chebyshev ratio tends to zero.

Weight	Why it is included
1	Unweighted baseline
$\log(i + 1)$	Slowly increasing logarithmic weight
$[\log(i + 1)]^2$	Stronger logarithmic growth
$1 + \log(i + 1) + [\log(i + 1)]^2$	Polynomial-log case
$\log \log(i + e)$	Extremely slowly increasing case

Table 1: Weights considered in 08A and 08B, in their common order.

1.2 Candidate weights

2 Positive log-power weights

The family $w_i = [\log(i + 1)]^p$, $p > 0$, provides one framework for different logarithmic growth rates. The exponent controls the relative emphasis on later validation observations. The question is whether the extra variability from increasing weights is dominated by the accumulated signal.

Theorem 2.1. *For every fixed $p > 0$,*

$$\text{Mean:} \quad \mathbb{E}[\Xi_n] \sim \frac{(\log n)^{p+1}}{p+1},$$

$$\text{Variance:} \quad \text{Var}(\Xi_n) \sim 2(\log n)^{2p},$$

and

$$\text{Selection:} \quad \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2} \sim \frac{2(p+1)^2}{(\log n)^2} \rightarrow 0.$$

Hence every fixed $p > 0$ gives selection consistency.

Weighting increases both signal and variance, but the squared mean grows two powers of $\log n$ faster. The full proof is given in Appendix B.

2.1 Two concrete choices

The log and squared-log weights are direct special cases.

Weight	$\mathbb{E}[\Xi_n]$	$\text{Var}(\Xi_n)$	Chebyshev ratio	Selection
$\log(i + 1)$	$\sim (\log n)^2/2$	$\sim 2(\log n)^2$	$\sim 8/(\log n)^2$	Consistent
$[\log(i + 1)]^2$	$\sim (\log n)^3/3$	$\sim 2(\log n)^4$	$\sim 18/(\log n)^2$	Consistent

Their finite-sample moments, ratios, and selection errors are examined in 08B, Sections 3–5.

3 Polynomial-log weights

The positive log-power result suggests that the dominant growth rate may determine the asymptotic behavior. This motivates a polynomial in $\log(i + 1)$.

Theorem 3.1. Suppose $w_i = P(\log(i+1)) > 0$, where $P(x) = cx^k + O(x^{k-1})$, $k \geq 1$, and $c > 0$. Then

$$\text{Mean:} \quad \mathbb{E}[\Xi_n] \sim \frac{c}{k+1} (\log n)^{k+1},$$

$$\text{Variance:} \quad \text{Var}(\Xi_n) \sim 2c^2 (\log n)^{2k},$$

and

$$\text{Selection:} \quad \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2} \sim \frac{2(k+1)^2}{(\log n)^2} \longrightarrow 0.$$

The highest-degree logarithmic term determines the leading behavior. Thus $1 + \log(i+1) + [\log(i+1)]^2$ has the same leading asymptotics as $[\log(i+1)]^2$. See Appendix C.1 for the proof and 08B, Sections 3–5 for the finite-sample comparison.

4 The log-log weight

The weight $w_i = \log \log(i+e)$ tests whether an even more slowly increasing sequence can still give a vanishing Chebyshev ratio.

Theorem 4.1. For $w_i = \log \log(i+e)$,

$$\text{Mean:} \quad \mathbb{E}[\Xi_n] \sim (\log n) \log \log n,$$

$$\text{Variance:} \quad \text{Var}(\Xi_n) \sim 2(\log \log n)^2,$$

and

$$\text{Selection:} \quad \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2} \sim \frac{2}{(\log n)^2} \longrightarrow 0.$$

Even this extremely slow increase is sufficient in the present example. The proof is in Appendix C.2, and the matching finite-sample evidence is in 08B, Sections 3–5.

5 Theory summary

Weight	$\mathbb{E}[\Xi_n]$	$\text{Var}(\Xi_n)$	Chebyshev ratio	Selection
1	$\sim \log n$	$\rightarrow \pi^2$	$\sim \pi^2/(\log n)^2$	Consistent
$\log(i+1)$	$\sim (\log n)^2/2$	$\sim 2(\log n)^2$	$\sim 8/(\log n)^2$	Consistent
$[\log(i+1)]^2$	$\sim (\log n)^3/3$	$\sim 2(\log n)^4$	$\sim 18/(\log n)^2$	Consistent
$1 + \log(i+1) + [\log(i+1)]^2$	$\sim (\log n)^3/3$	$\sim 2(\log n)^4$	$\sim 18/(\log n)^2$	Consistent
$\log \log(i+e)$	$\sim (\log n) \log \log n$	$\sim 2(\log \log n)^2$	$\sim 2/(\log n)^2$	Consistent

Table 2: Theoretical predictions examined in the accompanying simulation report.

6 Conclusion

Every fixed positive log-power weight considered here remains selection consistent because its accumulated signal dominates its variability. For polynomial-log weights, the highest-degree term determines the leading behavior. The log-log result shows that even extremely slow increasing growth can be sufficient.

The relevant property is not simply that the weights increase, but whether the accumulated weighted signal grows sufficiently fast relative to the standard deviation of Ξ_n . The accompanying 08B report checks how clearly these asymptotic predictions appear at finite sample sizes.

Appendix

A General mean and variance decomposition

Define $S_m = \sum_{t=1}^m \xi_t$. Then

$$\Xi_n = \sum_{i=2}^n w_i \left\{ \frac{S_{i-1}^2}{(i-1)^2} - 2\xi_i \frac{S_{i-1}}{i-1} \right\}. \quad (4)$$

Because S_{i-1} is independent of ξ_i , $\mathbb{E}(\xi_i S_{i-1}) = 0$ and $\mathbb{E}(S_{i-1}^2) = i-1$, proving (2).

For $1 \leq t < n$ and $2 \leq l \leq n$, define

$$a_{n,t} = \sum_{i=t+1}^n \frac{w_i}{(i-1)^2}, \quad b_{n,l} = \sum_{i=l+1}^n \frac{w_i}{(i-1)^2} - \frac{w_l}{l-1}.$$

Expanding S_{i-1}^2 and collecting Gaussian monomials gives

$$\Xi_n - \mathbb{E}[\Xi_n] = \sum_{t=1}^{n-1} a_{n,t} (\xi_t^2 - 1) + 2 \sum_{l=2}^n b_{n,l} \sum_{k < l} \xi_k \xi_l. \quad (5)$$

The centered squares have variance 2, and each product $\xi_k \xi_l$ has variance 1. Distinct terms are uncorrelated because some independent centered Gaussian variable appears to an odd power in every mixed product. There are $l-1$ products with larger index l . Hence

$$\text{Var}(\Xi_n) = 2 \sum_{t=1}^{n-1} a_{n,t}^2 + 4 \sum_{l=2}^n (l-1) b_{n,l}^2. \quad (6)$$

For the increasing weights below, define

$$A_t = \sum_{i=t+1}^{\infty} \frac{w_i}{(i-1)^2}, \quad B_l = A_l - \frac{w_l}{l-1}, \quad T_n = A_n.$$

Then $a_{n,t} = A_t - T_n$ and $b_{n,l} = B_l - T_n$ exactly. The leading terms in B_l cancel.

B Proof for increasing log-power weights

Proof of Theorem 2.1. Replacing $[\log(i+1)]^p/(i-1)$ by $(\log i)^p/i$ changes the mean by a bounded amount. With $u = \log x$, integral comparison gives

$$\mathbb{E}[\Xi_n] \sim \int^n \frac{(\log x)^p}{x} dx = \frac{(\log n)^{p+1}}{p+1}.$$

For $I_p(t) = \int_t^\infty (\log x)^p x^{-2} dx$, integration by parts gives $I_p(t) = (\log t)^p/t + p I_{p-1}(t)$. Substituting $x = te^v$ shows $I_q(t) = O((\log t)^q/t)$ for fixed q . Sum-integral comparison therefore yields

$$A_t \sim \frac{(\log t)^p}{t}, \quad T_n \sim \frac{(\log n)^p}{n}. \quad (7)$$

To expose the cancellation in B_l , replace $(i-1)^{-2}$ in A_l by $[i(i-1)]^{-1}$; the error is $O((\log l)^p/l^2)$. Summation by parts gives

$$\sum_{i=l+1}^{\infty} \frac{w_i}{i(i-1)} = \frac{w_{l+1}}{l} + \sum_{i=l+2}^{\infty} \frac{w_i - w_{i-1}}{i-1}.$$

The first term differs from $w_l/(l-1)$ by $O((\log l)^p/l^2 + (\log l)^{p-1}/l^2)$, and the mean-value theorem gives $|w_i - w_{i-1}| \leq C_p(\log i)^{p-1}/i$. Thus

$$|B_l| \leq C_p \left\{ \frac{(\log l)^{p-1}}{l} + \frac{(\log l)^p}{l^2} \right\}. \quad (8)$$

In particular, $\sum_t A_t^2 < \infty$, so the diagonal part of (6) is bounded.

For the nonterminal off-diagonal part,

$$\sum_{l=2}^n (l-1)B_l^2 = O\left(\sum_{l=2}^n \frac{(\log l)^{2p-2}}{l}\right).$$

With $u = \log x$, the comparison integral is $\int^{\log n} u^{2p-2} du$. It is bounded for $0 < p < 1/2$, of order $\log \log n$ for $p = 1/2$, and of order $(\log n)^{2p-1}$ for $p > 1/2$. In every case,

$$\sum_{l=2}^n (l-1)B_l^2 = o((\log n)^{2p}). \quad (9)$$

The terminal contribution satisfies

$$T_n^2 \sum_{l=2}^n (l-1) \sim \frac{(\log n)^{2p}}{2}.$$

After expanding $(B_l - T_n)^2$, Cauchy–Schwarz bounds the cross term by

$$\begin{aligned} 2 \left| T_n \sum_{l=2}^n (l-1)B_l \right| &\leq 2 \left\{ \left(\sum_{l=2}^n (l-1)B_l^2 \right) \left(T_n^2 \sum_{l=2}^n (l-1) \right) \right\}^{1/2} \\ &= o((\log n)^{2p}). \end{aligned}$$

The bounded diagonal part, the B_l^2 part, and the cross term are lower order. Therefore

$$\text{Var}(\Xi_n) \sim 4T_n^2 \sum_{l=2}^n (l-1) \sim 2(\log n)^{2p}.$$

Dividing by the squared mean completes the proof. $\square$

C Proofs for extensions

C.1 Polynomial-log weight

Proof of Theorem 3.1. Since $P(x) = cx^k + O(x^{k-1})$,

$$P(\log(i+1)) = c[\log(i+1)]^k + O([\log(i+1)]^{k-1}).$$

Integral comparison gives $\mathbb{E}[\Xi_n] = c(\log n)^{k+1}/(k+1) + O((\log n)^k)$. The tail argument above yields

$$A_t \sim \frac{c(\log t)^k}{t}, \quad B_l = O\left(\frac{(\log l)^{k-1}}{l} + \frac{(\log l)^k}{l^2}\right).$$

Hence $T_n \sim c(\log n)^k/n$, $\sum_{l=2}^n (l-1)B_l^2 = O((\log n)^{2k-1})$, and the cross term is lower order by Cauchy–Schwarz. The bounded diagonal term is also lower order, so

$$\text{Var}(\Xi_n) \sim 4T_n^2 \sum_{l=2}^n (l-1) \sim 2c^2(\log n)^{2k}.$$

□

C.2 Log-log weight

Proof of Theorem 4.1. Integral comparison and $u = \log x$ give

$$\mathbb{E}[\Xi_n] = \int^n \frac{\log \log x}{x} dx + O(1) = (\log n) \log \log n - \log n + O(1).$$

The same tail comparison and summation by parts give

$$A_t \sim \frac{\log \log t}{t}, \quad B_l = O\left(\frac{1}{l \log l} + \frac{\log \log l}{l^2}\right).$$

Both $\sum_t A_t^2$ and $\sum_l (l-1)B_l^2$ converge, while $T_n \sim \log \log n/n$. The cross term is lower order by Cauchy–Schwarz. Thus

$$\text{Var}(\Xi_n) \sim 4T_n^2 \sum_{l=2}^n (l-1) \sim 2(\log \log n)^2,$$

which proves the result. □